

Peripheral Devices (Gerce Birimleri)

MP
w311

- Giriş/Kıç cihazları
- Hafıza birimleri gibi belirli bir adrese yerleştirilir
- Adres genişliği dar

- Giriş birimi (in) hafızaya erişimle aynı
- Çıkış birimi (out) hafızaya erişimle aynı

• Assembly

I/O cihazları atomik olarak 8-bitlik

Adres
8-bit
Sığarsa

Fixed
Address

16-bit IN Komutu 16-bit OUT Komutu

8-bit IN AL, P8 8-bit OUT P8, AL

16-bit IN AX, P8 16-bit OUT P8, AX

$\overline{RD}$

M/I/O

ucaları
değişir

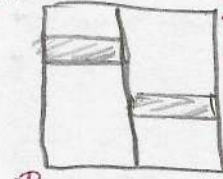
Adres
8-bit
Sığarsa

Var
Address

8-bit IN AL, DX 8-bit OUT DX, AL

16-bit IN AX, DX 16-bit OUT DX, AX

FFFF



I/O cihazları
adreslere
8-bit
veya
tek
ardışıl
olarak
ayarlanır!



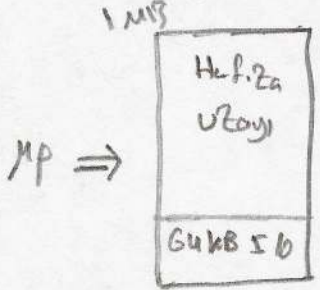
Memory erişiminde M/I/O=1, $\overline{RD}=0$, $\overline{WR}=0$ (Mov [offset])

Hafıza adres yolu genişliği 20-bit (Kapasite 1MB)

I/O adres yolu genişliği 16-bit (8086'da 64KB=2¹⁶ ayrılıyor)

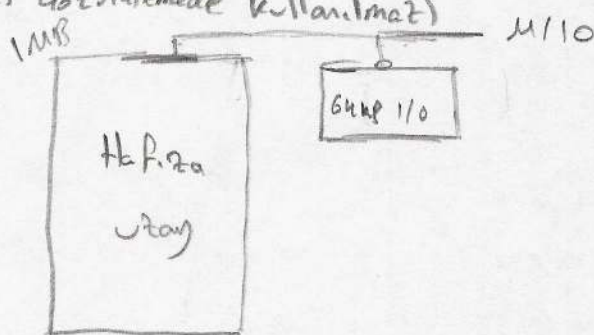
I/O Haritalama yöntemleri:

- Isolated I/O Mapping
- Memory Mapped I/O (M/I/O adres gözetiminde kullanılmaz)



Memory Mapped

- + Hafıza işlemlerini ayıran ve gerek yok
- + Özel komutlara gerek yok



Isolated I/O Mapping

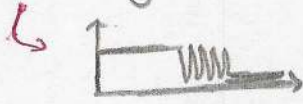
- + daha hızlı özel komutlar
- + Hafıza uçlarının tamamı hafıza birimleri için kullanılır
- + Basit adres gözetimleme

Dijital Arayüz

- Basılı iken ve bırakıldığında farklı lojik durum üretilmeli (Float olmamalı)

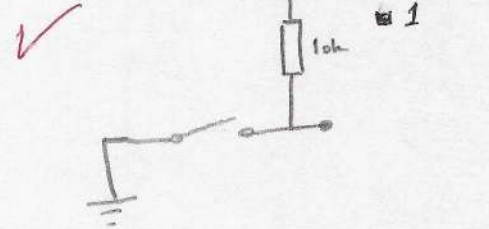
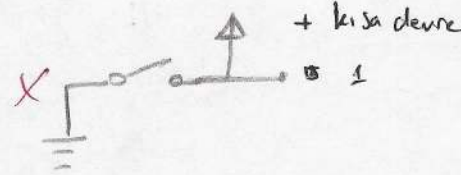
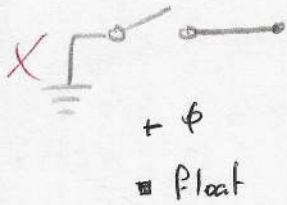
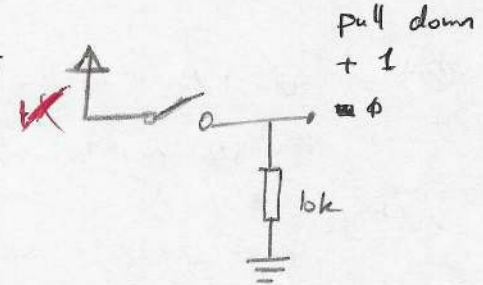
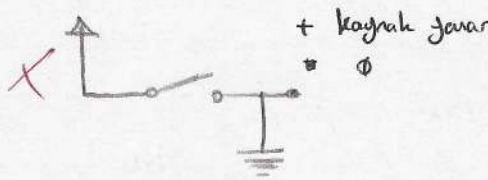
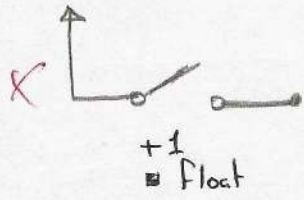
µp
CV3/2

- Kontakt gürültüsüne karşı yazılımsal veya donanımsal önlem alınmalıdır



Tasarımlar

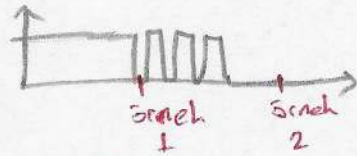
- + Basılı
- + Bırakıldı



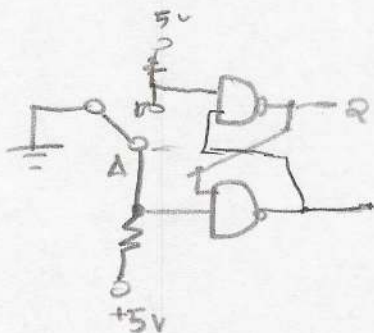
Kontakt Gürültüsü

* Yazılımsal Önlem

Üç okunduktan belli bir süre sonra tekrar okunur değer aynıysa doğru kabul edilir



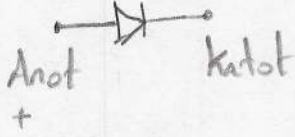
* Hardware Debounce Switch (Donanımsal Gözetim)



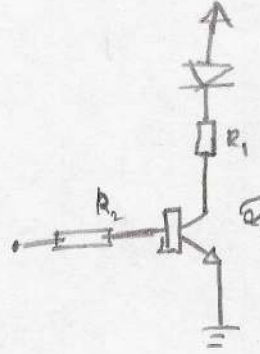
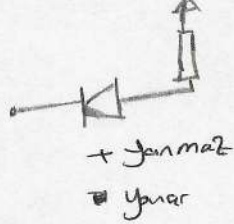
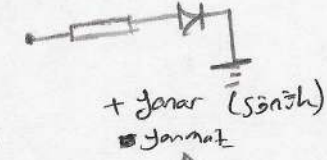
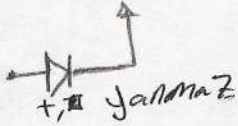
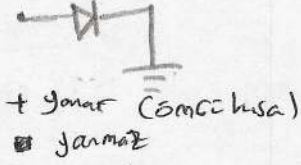
LED Arayışı

(Anot) katot $\Rightarrow$ yanar

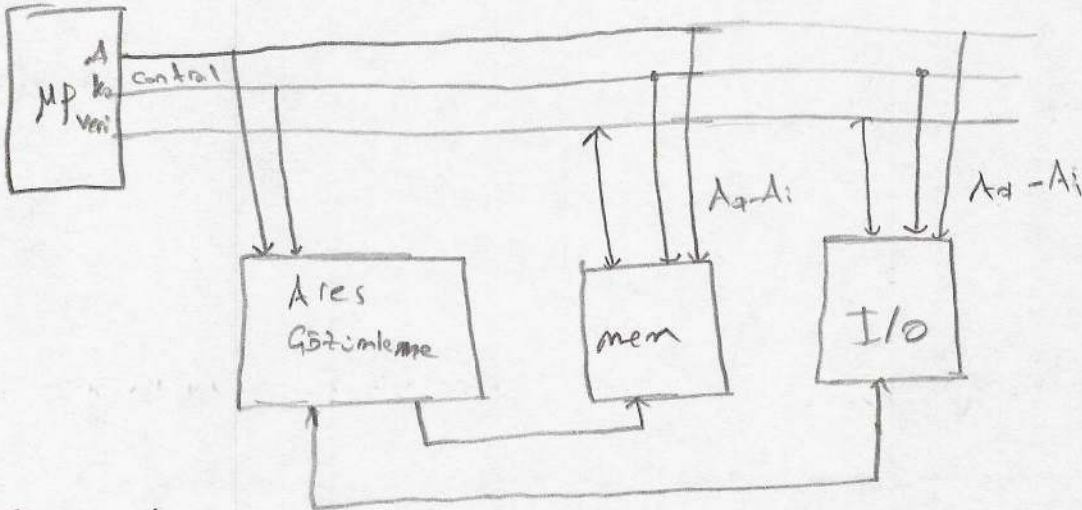
MP
W 3/3



Tasarımlar + Lohik 1
+ Lohik 0



Basit Çevre Birimleri için Adres Gözetimleme

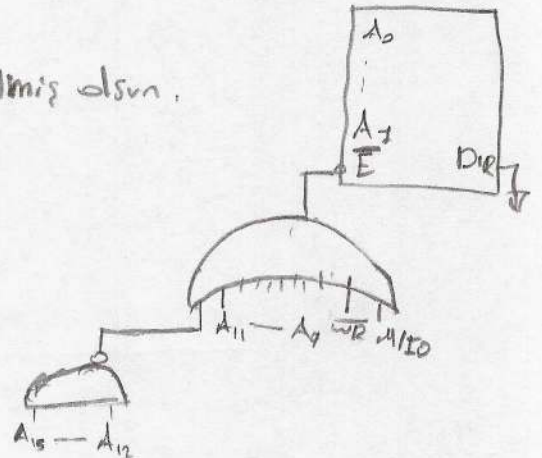


8086 için Veri nerede okur? (I/O)

Veri tipi Veri
Gıft adres (8bit) D0-D7
tek adres (8bit) D8-D15
Gıft adres (16bit) D0-D15

Ex. #4245 cihazı FFFFH adresine yerleştirilmiş olsun.

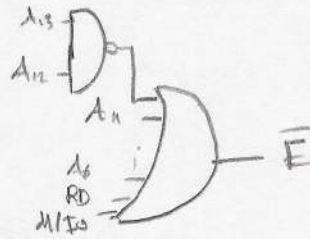
$\overline{WR} = 0$ $M/I/O = 0$



Ex. Basit Giriş Birimi (8-bitlik, F Çoğalt adresinde)

MP
W3/4

$\overline{RD} = \Phi$ $M/I\overline{O} = \Phi$

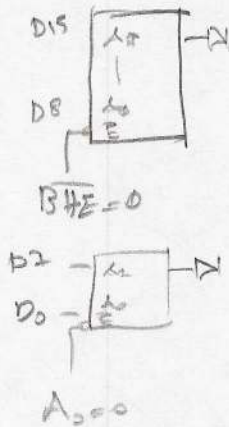


DIR'e Φ bağlanır

Mov cx, 8
LI: RCL AL, 1
Loop LI ; cx'te tuz no's var

RD ve WR uçları sayesinde aynı adrese giriş ve çıkış birimi bağlanabilir

Ex. 0F00H adresinden 16-bitlik çıkış



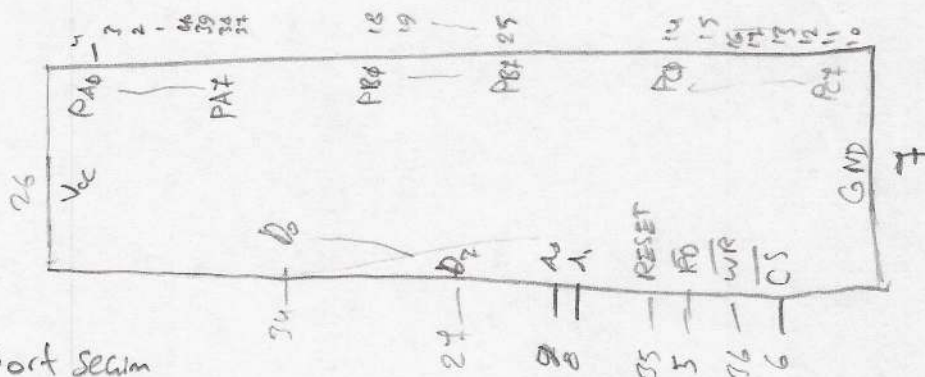
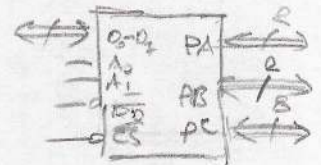
8086	A ₀	BHE
G1A 8bit	0	1
tek 8bit	1	0
G1A 16bit	0	0

8255 programlanabilir paralel veri I/O cihazı

Mod 0: Gönder, al ancak doğru bir şekilde çalıştığını kontrol etme

Mod 1: HandShaking

- 3 adet genel amaçlı port (8-bitlik)
- 8-bitlik veri yolu



- A₀, A₁ Port seçimi
- CS 8255 enable
- RD MP'ın okunması
- WR MP'nin veri çıkması

$A_4, A_3, \overline{RD}, \overline{WR}, \overline{CS}$

00 0 1 0 PA → Data Bus

01 0 1 0 PB → DB

10 0 1 0 PC → DB

11 0 1 0 Control word → DB

00 1 0 0 DB → PA

01 1 0 0 N → PB

10 1 0 0 PC → PC

11 1 0 0 / → Control word

MP
w3/5

İş yap

2 adet 8-bit PA, PB

2 adet 4-bit PC, PCH

16-bit Veriyolu 8-bitlik Çevre Birimi

$\overline{BHE}, A_4$

0 0 Antamlı değil

0 1 Tek adreste byte D8-D15

1 0 8-bit ~ ~ D0-D7

1 1 Antamlı değil

Ex. 8255 PPI with starting address of Feh. only even addresses!

$A_7=1$

$A_6=1$

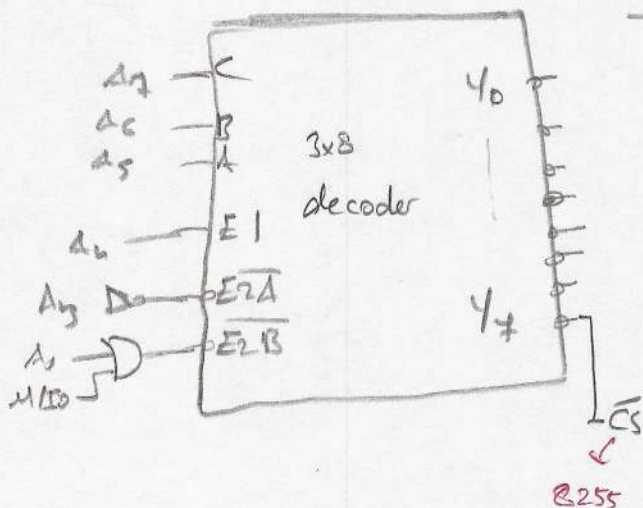
$A_5=1$

$A_4=\phi$

$M/\overline{IO}=\phi$

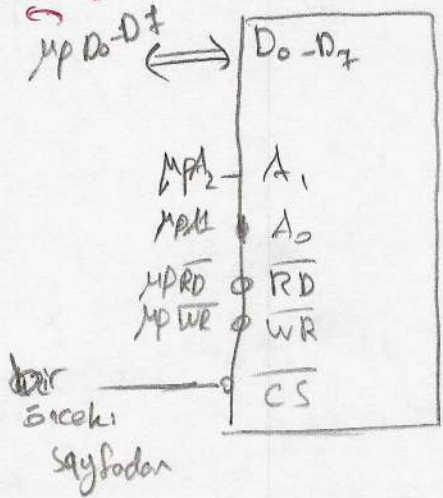
	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
F8H	1	1	1	1	1	0	0	0
FAH	1	1	1	1	1	0	1	0
FCH	1	1	1	1	1	1	0	0
FEH	1	1	1	1	1	1	1	0

A_2-A_1
Register
Select



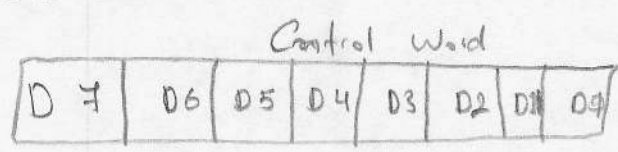
E_1	$\overline{E_2A}$	$\overline{E_2B}$	C	B	A	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	x	x	x	x	x	1	1	1	1	1	1	1	1
x	1	x	x	x	x	1	1	1	1	1	1	1	1
x	x	1	x	x	x	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	0	1	0	1	0	1	1	1	1	1	1
1	0	0	0	1	1	1	0	0	1	1	1	1	1
1	0	0	1	0	0	1	0	0	1	1	1	1	1
1	0	0	1	0	1	1	0	0	0	1	1	1	1
1	0	0	1	1	0	1	0	0	0	0	1	1	1
1	0	0	1	1	1	1	0	0	0	0	0	1	1

MP Adres oldu için 8255



MP
WSIG

8255 - Mod 0



D7: Mode Set Flag 1 = active

Group A

D6-D4: 00: mod 0, 01: mod 1, 1x: mod 2

D4: 0: PA çıkış, 1: PA giriş

D3: 0: PC_(upper) çıkış, 1: PC_(upper) giriş

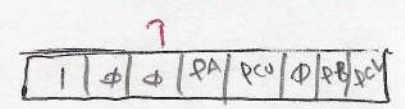
Group B

D2: 0: mod 0, 1: mod 1

D1: 0: PB çıkış, 1: PB giriş

D0: 0: PC_(lower) çıkış, 1: PC_(lower) giriş

Tüm portlarda mod 0



• Tüm portlar çıkış
CW ← 86H

• Tüm portlar giriş
CW ← 9BH

CODE SEGMENT PUBLIC 'CODE'

ASSUME CS: CODE

START:

endless:

out 62H, AL

Jump endless

CODE ENDS

END START

62H'taki porta

veri çıkarmayı deniyor